The Beatty logarithm bound by $\pi^2/6$

Principia Math In Progress

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Task 1 (MathDB 330197). Polanco's Conjecture 1 [1] asks whether, for all irrational $a, b > 1$,

$$\left|\log(a/b) - \sum_{n=1}^{\infty} \frac{a\,\{(n+1)/a\} - b\,\{(n+1)/b\}}{n(n+1)}\right| < \frac{\pi^2}{6}.$$

We prove this directly from a finite summation identity.

Theorem 1. *For irrational $a > 1$, define*

$$R(a) = \sum_{m=1}^{\infty} \left(\frac{a}{\lfloor am \rfloor} - \frac{1}{m}\right).$$

Then $0 < R(a) < \pi^2/6$ and

$$a \sum_{n=1}^{\infty} \frac{\{(n+1)/a\}}{n(n+1)} = \log a + 1 - R(a). \tag{1}$$

Consequently the expression inside the conjectured absolute value is $R(a) - R(b)$, and the conjecture holds strictly.

Proof. Since $\lfloor am \rfloor \geq m$ and $0 < \{am\} < 1$,

$$0 < \frac{a}{\lfloor am \rfloor} - \frac{1}{m} = \frac{\{am\}}{m\lfloor am \rfloor} < \frac{1}{m^2}. \tag{2}$$

The series therefore converges and is positive. Its sum is strictly less than $\sum_{m \geq 1} m^{-2} = \pi^2/6$: preserve the positive gap in the first term and use a non-strict upper bound for the rest.

Let $N \geq 1$ and $M = \lfloor (N+1)/a \rfloor$. Because a is irrational, for positive integers m, n,

$$m \leq \left\lfloor \frac{n+1}{a} \right\rfloor \quad \Longleftrightarrow \quad am < n+1 \quad \Longleftrightarrow \quad \lfloor am \rfloor \leq n.$$

All sums in the following interchange are finite. Telescoping gives

$$\begin{aligned} \sum_{n=1}^{N} \frac{\lfloor (n+1)/a \rfloor}{n(n+1)} &= \sum_{m=1}^{M} \sum_{n=\lfloor am \rfloor}^{N} \frac{1}{n(n+1)} \\ &= \sum_{m=1}^{M} \left(\frac{1}{\lfloor am \rfloor} - \frac{1}{N+1}\right). \end{aligned}$$

Let $H_j = \sum_{n=1}^{j} 1/n$, with $H_0 = 0$, and let $R_M(a)$ denote the first M terms of $R(a)$. Expanding the fractional parts gives

$$a \sum_{n=1}^{N} \frac{\{(n+1)/a\}}{n(n+1)} = H_N - H_M - R_M(a) + \frac{aM}{N+1}. \tag{3}$$

As $N \to \infty$, we have $M \sim N/a$. Harmonic-number asymptotics therefore give $H_N - H_M \to \log a$. Also $aM/(N+1) \to 1$ and $R_M(a) \to R(a)$. The series on the left converges absolutely by comparison with $a/[n(n+1)]$. This proves (1). Applying it to a and b cancels the logarithms and constants. Finally, the distance between two numbers strictly inside $(0, \pi^2/6)$ is strictly less than $\pi^2/6$. □

Effective remainder. For every integer $L \geq 1$, (2) also gives

$$0 < R(a) - R_L(a) < \sum_{m>L} \frac{1}{m^2} < \frac{1}{L}.$$

Thus the error in replacing $R(a) - R(b)$ by $R_L(a) - R_L(b)$ is less than $1/L$. This is an analytic error bound, not a numerical assumption.

Scope and verification. Conjecture 1 appears on page 4 of arXiv:1503.08512v1 and on journal page 104 of the final article. Both versions allow arbitrary irrational $a, b > 1$. Our proof establishes the signed identity directly from the defining series and does not require the source's other identities. The separate bound by one is false, as shown in the distinct package for MathDB 330198. Exact checks of (3) accompany this proof; the argument above covers all irrational parameters and all indices.

Proposed follow-up tasks (not claimed solved).

2. Determine the sharp value of $\sup_{a,b>1,\ a,b\notin\mathbb{Q}} |R(a) - R(b)|$.
3. Derive the corresponding finite summation and limiting formulas when $1/[n(n+1)]$ is replaced by $1/[n(n+r)]$ for a fixed integer $r > 1$.
4. Obtain parameter-dependent upper bounds for $R(a)$ that improve $\pi^2/6$ and remain effective uniformly on compact subintervals of $(1, \infty)$.
5. Extend (1) to rational $a > 1$, accounting explicitly for all indices at which $(n+1)/a$ is integral.

References

[1] G. Polanco, *The logarithm of irrational numbers and Beatty sequences*, Acta Arithmetica **179** (2017), 101–123, doi:10.4064/aa6695-1-2017. Preprint: arXiv:1503.08512v1 (2015).